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Appendix A. Implementation Details

The LTN library is implemented in Tensorflow 2 [1] and is available from GitHub[29]. Every logical operator is grounded using Tensorflow primitives. The LTN code implements directly a Tensorflow graph. Due to Tensorflow built-in optimization, LTN is relatively efficient while providing the expressive power of FOL.

Table A.3 shows an overview of the network architectures used to obtain the results of the examples in Section 4. The LTN repository includes the code for these examples. Except if explicitly mentioned otherwise, the reported results are averaged over 10 runs using a 95% confidence interval. Every example uses a stable real product configuration to approximate Real Logic operators, and the Adam optimizer [35] with a learning rate of 0.001 to train the parameters.

Task	Network	Architecture
4.1	`MLP`	Dense(16)*, Dense(16)*, Dense(1)
4.2	`MLP`	Dense(16)*, Dropout(0.2), Dense(16)*, Dropout(0.2), Dense(8)*, Dropout(0.2), Dense(1)
4.3	`MLP`	Dense(16)*, Dense(16)*, Dense(8)*, Dense(1)
4.4	`CNN`	MNISTConv, Dense(84)*, Dense(10)
	baseline − SD	MNISTConv × 2, Dense(84)*, Dense(19), Softmax
	baseline − MD	MNISTConv × 4, Dense(128)*, Dense(199), Softmax
4.5	`MLP`	Dense(8)*, Dense(8)*, Dense(1)
4.6	`MLP`	Dense(16)*, Dense(16)*, Dense(16)*, Dense(1)
4.7	`MLP_S`	Dense(8)*, Dense(8)*, Dense(1)
	`MLP_F`	Dense(8)*, Dense(8)*, Dense(1)
	`MLP_C`	Dense(8)*, Dense(8)*, Dense(1)

* : layer ends with an `elu` activation
Dense(n) : regular fully-connected layer of n units
Dropout(r) : dropout layer with rate r
Conv(f, k) : 2D convolution layer with f filters and a kernel of size k
MP(w, h) : max pooling operation with a $w \times h$ pooling window
MNISTConv : Conv(6, 5)*, MP(2, 2), Conv(16, 5)*, MP(2, 2), Dense(100)*

Table A.3: Overview of the neural network architectures used in each example. Notice that in the examples, the networks are usually used with some additional layer(s) to ground symbols. For instance, in experiment 4.2, in $\mathcal{G}(P) : x, l \mapsto l^\top \mathtt{softmax}(\mathtt{MLP}(x))$, the softmax layer normalizes the raw predictions of `MLP` to probabilities in [0, 1], and the multiplication with the one-hot label l selects the probability for one given class.

[29] https://github.com/logictensornetworks/logictensornetworks